

1. Create a simple JavaScript function called `calculateTotal` that takes two numbers: `itemPrice` and `quantity`, and returns the total bill amount using arithmetic operators.

```
function calculateTotal(itemPrice, quantity) {  
    return itemPrice * quantity;  
}
```

```
let total = calculateTotal(500, 3);
```

```
console.log("Total Bill: ₹" + total);
```

Output:

Total Bill: ₹1500

2. Build a Flipkart-style discount calculator: given product price, discount percentage, and a boolean `isMember`, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if `isMember` is true).

```
function calculateDiscount(price, discount, isMember) {  
    let discountAmount = price * discount / 100;  
    let finalPrice = price - discountAmount;  
  
    if (isMember == true) {  
        finalPrice = finalPrice - (finalPrice * 5 / 100);  
    }  
  
    return finalPrice;  
}
```

```
let price = 2000;
```

```
let discount = 10;
```

```
let isMember = true;
```

```
let finalPrice = calculateDiscount(price, discount, isMember);
```

```
console.log("Final Price: ₹" + finalPrice);
```

Output:

Final Price: ₹1710

Here:

- 10% discount → ₹1800
- Extra 5% member discount → ₹1710

3. Write a function `isEligibleForOffer` that takes a user's age and total order value, and returns true if the user is 18 or older AND the order value is above 500, otherwise false.
Hint: Use relational and logical operators together.

```
function isEligibleForOffer(age, orderValue) {  
    return age >= 18 && orderValue > 500;  
}
```

```
console.log(isEligibleForOffer(20, 700));
```

```
console.log(isEligibleForOffer(16, 700));
```

Output:

true

false

&& means **AND**, so both conditions must be true.

4. Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.

Condition:

At least 1000 likes OR (more than 200 comments AND at least 50 shares)

```
let likes = 1200;
```

```
let comments = 150;
```

```
let shares = 60;
```

```
let trending = likes >= 1000 || (comments > 200 && shares >= 50);
```

```
console.log("Is the post trending?", trending);
```

Output:

Is the post trending? true

- `||` → OR
- `&&` → AND
- `>=` → Greater than or equal to
- `>` → Greater than

5. Write a code snippet that demonstrates the difference between pre-increment (`++count`) and post-increment (`count++`) by logging the values before and after using both on a `followerCount` variable.

```
let followerCount = 100;
```

```
console.log("Before:", followerCount);
```

```
console.log("Post-increment:", followerCount++);
```

```
console.log("After post-increment:", followerCount);
```

```
console.log("Pre-increment:", ++followerCount);  
console.log("After pre-increment:", followerCount);
```

Output:

Before: 100

Post-increment: 100

After post-increment: 101

Pre-increment: 102

After pre-increment: 102

Easy Difference

- `count++` → **First use the value, then increase it.**
- `++count` → **First increase the value, then use it.**